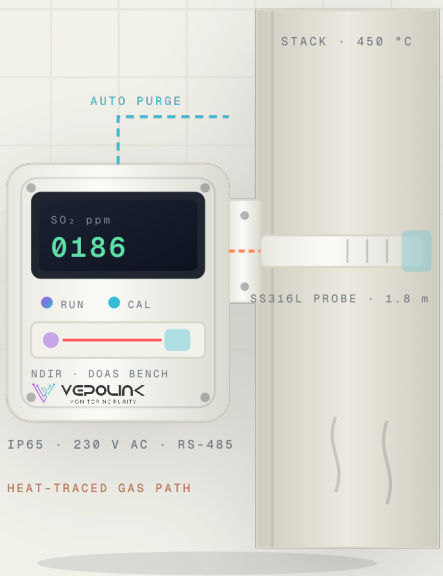
VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × EMISSION MONITORING

IN-SITU FLUE / STACK ANALYZER · NDIR & DOAS

In-Situ Flue / Stack Gas Analyzer *One flange on the stack*

A highly integrated single-flange gas monitoring system that measures CO, CO_2, SO_2, NO_2, NOx and O_2 directly inside the pipeline — heat-traced, auto-purged and built for dusty flue gas up to 450 °C.



450 °C
OPERATING TEMPERATURE

≤10 s
DETECTOR RESPONSE TIME

1.8 m
SS316L IN-STACK PROBE

IP65 DUST PROOF | **AUTO** ZERO & SPAN | **RS-485** OUTPUT

2%	±2% FS	2%/mo	≤10s
ACCURACY OF MEASURED VALUE	REPEATABILITY	ZERO & SPAN DRIFT	DETECTOR RESPONSE

01 OVERVIEW

Measured inside *the pipeline.*

A highly integrated single-flange gas monitoring system for accurate online measurement of industrial stack and flue gases, using NDIR and DOAS absorption to read trace concentrations directly inside the pipeline.

The probe carries the gas path; the analyzer body carries the optical bench, the filter and the purge. Heat tracing runs the full length of the gas path so nothing condenses at 450 °C flue temperature, and an automatic purge keeps dust and salt from blocking the probe. It suits boiler flues, denitrification systems, exhaust monitoring and general industrial emission monitoring.

AT A GLANCE

- NDIR
- DOAS
- SINGLE FLANGE
- HEAT TRACED
- AUTO PURGE
- SS316L PROBE
- UP TO 450 °C
- IP65

· PRODUCT FEATURES

Single-Flange Installation

The whole system mounts on one flange — probe, gas path, optical bench and purge in a single integrated assembly. No opposed mounting, no sample shelter.

Hot Gas Path, Kept Hot

High-temperature heat tracing throughout the gas path prevents condensation and the loss of soluble gases, with operation up to 450 °C flue temperature.

Automatic Purge

Scheduled purging clears dust and salt before they block the probe, which is what makes the system workable in heavily dust-laden flues.

Self-Calibrating, Serviceable

Manual and automatic zero and span calibration, a digital microcontroller display, easy-to-replace filter elements and a high-temperature flowmeter on the gas path.

02 WORKING PRINCIPLE

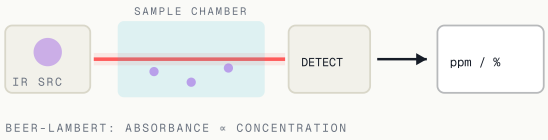
Light through gas, *read as concentration.*

An infrared source passes light through the sample gas chamber toward a detector. Gas molecules absorb specific wavelengths according to the Beer-Lambert law, and the resulting attenuation gives the concentration of the selected gas.

NDIR

Infrared Absorption

CO · CO_2 · SO_2 · NO_2



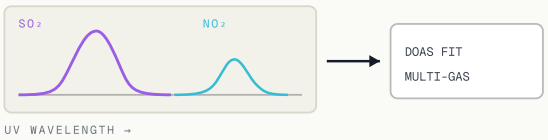
Non-dispersive infrared measures how much light each component removes at its own absorption band. The attenuation is converted directly into concentration for the selected gas.

`Absorbance ∝ concentration`

DOAS

Differential Optical Absorption

MULTI-GAS RESOLUTION IN ONE SPECTRUM



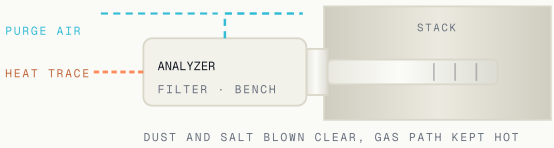
DOAS separates the narrow structure of each species from broad background attenuation, so overlapping components are resolved and dust or drift in the light path does not bias the result.

`Structure, not just intensity`

GAS PATH

Probe, Filter, Purge

SS316L 1.8 M · HEAT TRACED · FLOWMETER



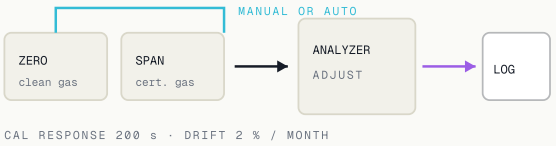
The stainless probe draws flue gas through replaceable filter elements into the measuring chamber, with a high-temperature flowmeter monitoring flow and an automatic purge clearing dust and salt.

`Hot, clean, monitored`

CALIBRATION

Manual or Automatic

ZERO & SPAN · 200 S CALIBRATION RESPONSE



Zero and span can be run by hand or on an automatic cycle. Drift is 2 % per month on both, so scheduled calibration keeps the instrument comfortably inside specification.

`Verified on a schedule`

03 MEASURING RANGES

Ranges and *configuration.*

Standard ranges for the common flue components. Other gases and ranges are configured to the application and the consent limits of the installation.

GAS	NAME	MEASURING RANGE
CO	Carbon Monoxide	0 – 3000 ppm
CO_2	Carbon Dioxide	0 – 25 %
SO_2	Sulphur Dioxide	0 – 3000 ppm

GAS	NAME	MEASURING RANGE
NO_2	Nitrogen Dioxide	0 – 1500 ppm
NOx	Nitrogen Oxides	As configured
O_2	Oxygen	As configured

STEP 01

Flange

A single flange on the stack carries the 1.8 m SS316L probe and the analyzer body — one opening, one assembly.

STEP 02

Heat trace

The gas path is heat traced end to end so flue gas stays above dew point and soluble components are not lost.

STEP 03

Measure

NDIR and DOAS benches read the components in the measuring chamber, with flow monitored by a high-temperature flowmeter.

STEP 04

Purge & calibrate

Automatic purge clears dust and salt; zero and span run manually or automatically, with a 200 s calibration response.

Because measurement happens at the stack, there is no shelter, no long heated line and no sample pump to keep alive. The instrument is designed for strong adaptability to changing operating conditions — the load swings, the dust rises, and the reading continues.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

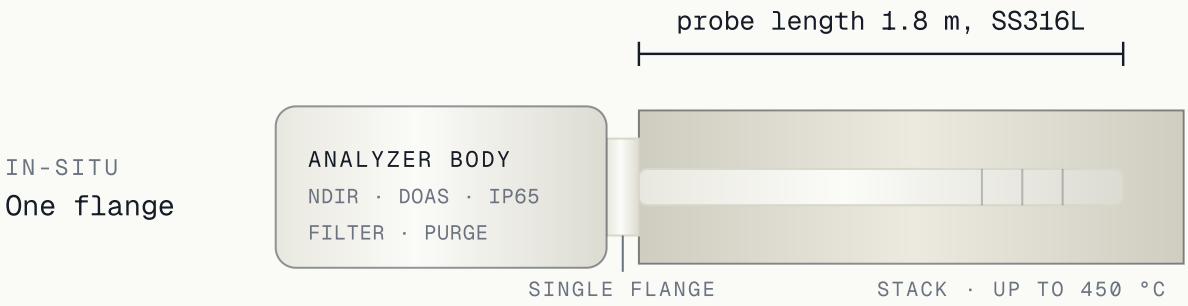
Principle	NDIR, with DOAS optical absorption
Measurement Type	In-situ, inside the pipeline
Measured Gases	CO, CO_2, SO_2, NO_2, NOx, O_2
Accuracy	2 % of measured value
Repeatability	±2 % F.S.
Zero Drift	2 % per month
Span Drift	2 % per month
Detector Response Time	≤ 10 s
Calibration Response Time	200 s
Calibration	Manual & automatic zero and span

BUILD · UTILITIES

Installation	Single-flange, in-situ
Probe Material	SS316L
Probe Length	1.8 m
Operating Temperature	Up to 450 °C
Gas Path	High-temperature heat tracing throughout
Flow Monitoring	High-temperature flowmeter
Filtration	Easy-to-replace filter elements
Purge	Automatic, against dust and salt blockage
Display	Digital, microcontroller based
Digital Output	1 × RS-485
Power Supply	230 V AC ±10 %, 50 Hz
Analyzer Body	IP65 dust proof

PROBE GEOMETRY

SINGLE FLANGE · 1.8 M SS316L PROBE IN THE FLUE



05 INSTALLATION & UPKEEP

One opening, *one assembly.*

> In dusty flues the instrument lives or dies by three things: the **filter elements**, the **purge schedule** and the **heat tracing**. Keep those right and the system holds its 2 % monthly drift; neglect them and no optical bench will save the reading.

01 · LOCATION

Representative section

A straight run of flue with well-mixed flow, away from bends, dampers and air ingress, within reach of a service platform.

02 · FLANGE

Single opening

One flange carries the probe and the analyzer body. Nothing is mounted on the far wall and no shelter is required.

03 · PROBE

1.8 m SS316L

Stainless probe reaches into the flow; replaceable filter elements take the dust before the measuring chamber.

04 · HEAT TRACE

Full gas path

High-temperature tracing keeps the whole path hot, so nothing condenses and soluble gases arrive intact at the bench.

05 · PURGE

On a schedule

Automatic purging clears dust and salt from the probe, which is what allows the system to work in heavy-dust service.

06 · CALIBRATION

Zero and span

Run by hand or automatically, with a 200 s calibration response. Filter elements are changed on the same visit.

IN-SITU VERSUS EXTRACTIVE

An extractive CEMS brings the gas to a conditioned cabinet and is usually the choice where many components must be reported with full traceability. This single-flange in-situ analyzer removes the heated line, the pump and the shelter entirely — a faster, cheaper installation where a handful of components is enough and the flue is hot and dusty.

INTEGRATION

An RS-485 digital output carries values to industrial emission monitoring and data acquisition systems. Paired with a Vepolink IoT data logger, the same stream feeds your SCADA and the CPCB or SPCB endpoint without a second instrument.

06 APPLICATIONS

Where it *works.*

Hot, dusty flues where the measurement has to happen at the stack rather than in a shelter.

Industrial Emission Monitoring	Continuous Emissions Monitoring	Boiler Flue Monitoring
Denitrification Systems	Exhaust Gas Monitoring	Power Plants
Cement Plants	Manufacturing Industries	Industrial Process Monitoring

DENITRIFICATION CONTROL

On an SCR or SNCR system the analyzer is not only a reporting instrument: NOx before and after the reactor governs reagent dosing, and a ten-second detector response is fast enough for that loop. Correct dosing is the difference between meeting the limit and paying for reagent you did not need.

SPECIFYING A POINT

We need the components and ranges you must report, the flue temperature and dust loading, the wall thickness and available flange, and the access arrangement for service. From that we confirm probe, filtration, purge, calibration and output as one package.

07 FAQ

Frequently asked *questions.*

Q What is the in-situ stack gas analyzer?

An online in-situ analyzer for continuous measurement of industrial flue and stack emissions, measuring inside the pipeline rather than in a shelter.

Q Which gases can it measure?

CO, CO_2, SO_2 and NO_2 as standard, with NOx and O_2 configurable according to application requirements.

Q How does it work?

NDIR and DOAS optical absorption. Infrared and UV light crossing the sample chamber is attenuated by the target gas, and the attenuation gives concentration.

Q Where is it used?

Boiler flues, denitrification systems, exhaust stacks, power plants, cement plants and other industrial emission monitoring applications.

Q What are its main features?

High-temperature operation, digital display, single-flange installation, heat-traced gas path, automatic purge and manual or automatic zero and span calibration.

Q How hot can the flue be?

Up to 450 °C operating temperature, with heat tracing along the full gas path to prevent condensation.

Q How does it cope with dust?

Easy-to-replace filter elements protect the measuring chamber and an automatic purge system clears dust and salt before they block the probe.

Q What is the probe made of?

SS316L stainless steel, 1.8 m long, suited to corrosive flue conditions at high temperature.

Q How is data taken out?

Through one RS-485 digital output, for integration with industrial emission monitoring and data acquisition systems.

Q How stable is it between calibrations?

Zero and span drift are 2 % per month, with 2 % accuracy of measured value and ±2 % F.S. repeatability.

This brochure is general product information. Measured components, ranges, probe configuration and performance figures are indicative and confirmed at order against the flue conditions and reporting requirements of the installation. Siting, calibration and reporting must follow the applicable national standards and the consent conditions of the site. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

MEASURED AT THE STACK

One flange, *and you're monitoring.*

Send us the components and ranges you report, the flue temperature and dust loading, and the flange you have — we'll confirm probe, filtration, purge and output as one package. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specs, installs and calibrates stack analyzers. No call centre.
+91 99538 86178 →

02 **Request a configuration** RECOMMENDED
Share gases, ranges, flue temperature, dust loading and flange details — we'll return the full specification.
sales@vepolink.com →

03 **Book a demonstration**
See a live analyzer, an automatic purge and calibration cycle, and data on an RS-485 link.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM